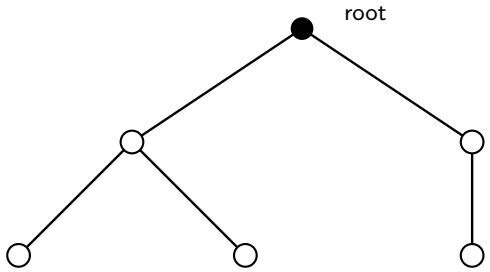
Week 12 / Catalan bijections / Grades 4–5

Draw dots joined by lines called edges. The filled dot at the top is the root. Each edge joins a parent dot to a child dot below it. Every dot except the root has exactly one parent. Keep each parent's children in a fixed left-to-right order. All dots connect to the root. Only the branching and its order matter.

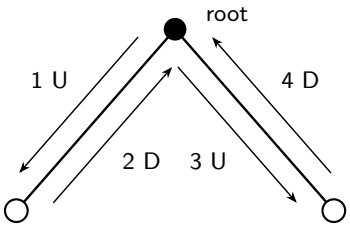
Example with five edges:



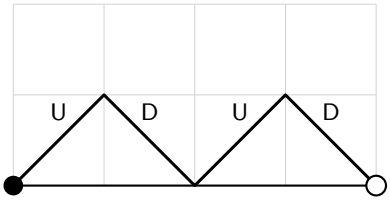
Problem 1: Draw all different trees with three edges, using these rules. How do you know your collection is complete?

Start and finish at the root. Visit each child's whole branch from left to right before visiting the next child. Write U whenever you go away from the root along an edge and D whenever you return along it.

Example:

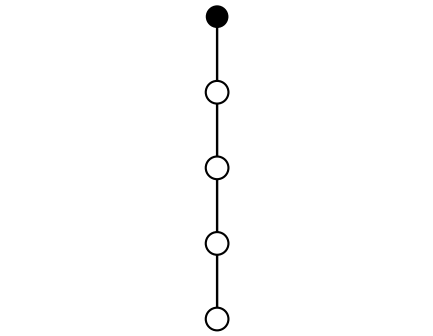


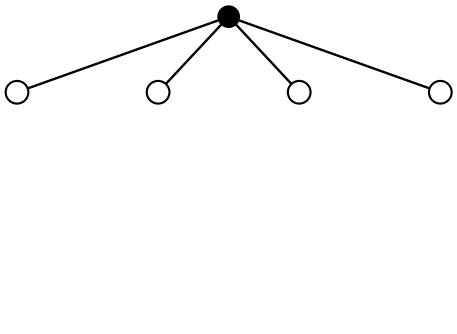
On the path, U goes up and D goes down.

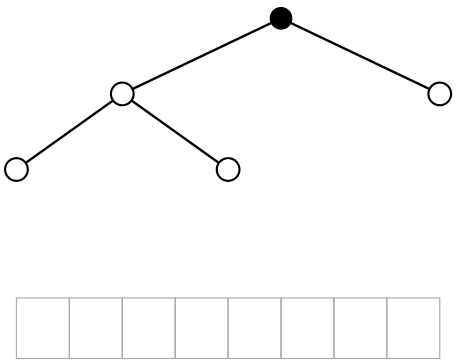


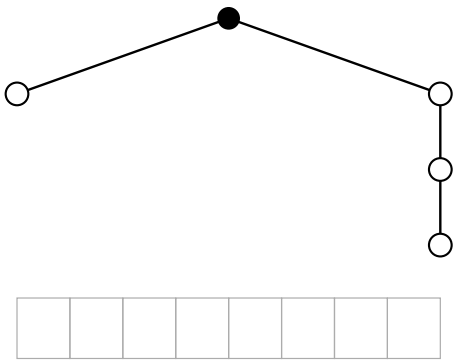
Walk code: `UDUD`

Problem 2: Write the walk code for each tree. Which features of a tree can you read from its code?









Problem 3: Draw a tree for each code. Can a code describe two different trees?

UUDDUD



UDUUDD



UUDUDUDD



UUUDDUDD



Problem 4: Which of these codes could come from a tree? Give rules that decide for any string of U's and D's, and explain why they work. Describe how to rebuild a tree from any code that follows your rules.

UUUUDDDD

UUDDDUUD

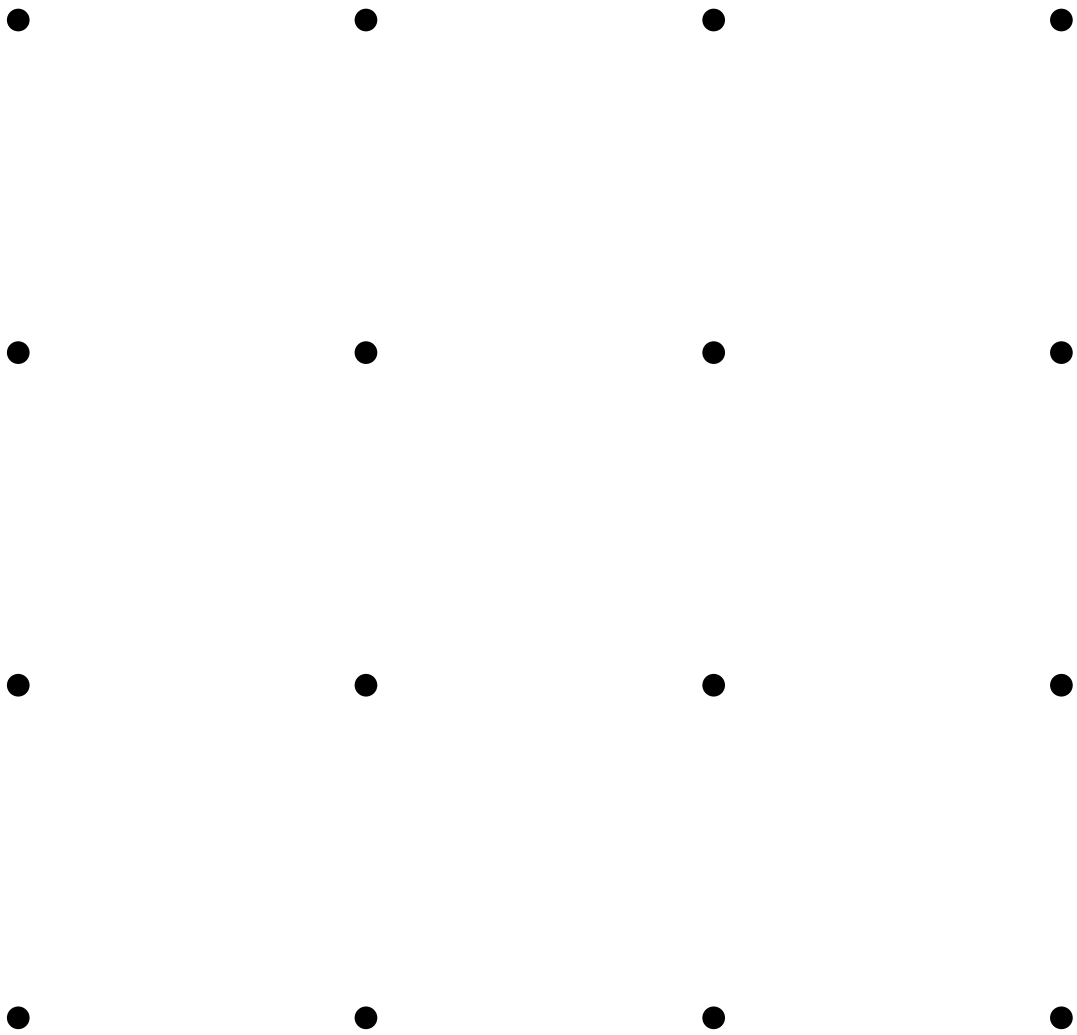
UDUDUUDD

UUDUDUDU

UUUDUDDD

DUUUDDUD

Problem 5: How many different trees have four edges? Organize the count so someone can check that every tree is counted exactly once.



Problem 6: Find the number of trees with five edges, then six edges. Give a counting rule that works for any number of edges, and explain why it counts each tree once.